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ECON1050 Tutorial 7

Partial Derivatives and Multivariate Optimisation

Week 9 - Semester 1 2026

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Today's Plan

- Recap: key ideas from Lecture 7
- Tutorial questions: Q1–Q4

Recap: Functions of two variables

Definition

A function f of two real variables x and y with domain D is a rule that assigns a specified number $f(x, y)$ to each point (x, y) in D.

Notation and interpretation

$$z = f(x, y).$$

$$\begin{matrix} y \rightarrow \\ x \rightarrow \end{matrix} \boxed{f(\,)} \rightarrow z$$

- x and y: independent or exogenous variables; arguments of f.
- z: dependent or endogenous variable.
- Domain of f: all possible pairs (x, y).
- Range of f: corresponding values of z.

Recap: Partial derivatives

Core idea

Partial derivatives measure how a function changes when one variable changes while the other variables are held constant.

x y

Notation

For $z = f(x, y)$,

f_x f_1 f_y f_2

$$\frac{\partial z}{\partial x} = f_x(x, y) = f_1(x, y), \qquad \frac{\partial z}{\partial y} = f_y(x, y) = f_2(x, y).$$

- f_x: rate of change with respect to x, holding y constant.
- f_y: rate of change with respect to y, holding x constant.

Derivative $f'(x)$: rate of change

$\Delta x \to f(x)$

$\Delta x \to f(x,y)$

$\Delta y \to$

Recap: Higher-order partial derivatives

Second-order partial derivatives

For a function $f(x, y)$, there are four second-order partial derivatives:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \qquad f_{yy} = \frac{\partial^2 f}{\partial y^2},$$

$$f_{xy} = \frac{\partial^2 f}{\partial y \partial x}, \qquad f_{yx} = \frac{\partial^2 f}{\partial x \partial y}.$$

Mixed partials theorem

When the relevant derivatives are continuous, the mixed partials are equal:

$$f_{xy} = f_{yx}.$$

$$f_{xy} = f_{yx}$$

Recap: Economic interpretation of cross effects

Cross-price effects

If demand for good A depends on the price of good B, then

$$\boxed{\frac{\partial q_A}{\partial p_B}}$$

measures how demand for A responds to the price of B.

Substitutes vs complements

- If $\frac{\partial q_A}{\partial p_B} > 0$, then A and B are substitutes.
- If $\frac{\partial q_A}{\partial p_B} < 0$, then A and B are complements.
- If $\frac{\partial q_A}{\partial p_B} = 0$, there is no cross-price effect.

Recap: Multivariate optimisation and FOCs

Interior stationary points

For a differentiable function $z = f(x, y)$, an interior maximum or minimum can occur only at a stationary point.

First-order conditions

At an interior stationary point (x_0, y_0),

$f_x = 0$ and $f_y = 0$

$$f_x(x_0, y_0) = 0, \qquad f_y(x_0, y_0) = 0.$$

Equivalently, the gradient vector is zero:

$$\nabla \boldsymbol{f}(x_0, y_0) = \begin{pmatrix} f_x(x_0, y_0) \\ f_y(x_0, y_0) \end{pmatrix} = \mathbf{0}.$$

Recap: Second-order conditions for local extrema

Hessian matrix

At an interior stationary point (x_0, y_0), define

$$\boldsymbol{H}(x_0, y_0) = \begin{pmatrix} A & B \\ B & C \end{pmatrix}, \quad A = f_{xx},\ B = f_{xy},\ C = f_{yy}.$$

$A = f_{xx}$ $B = f_{xy}$ and f_{yx} $C = f_{yy}$

Let

$$D = AC - B^2.$$

$= f_{xx} \cdot f_{yy} - f_{xy} \cdot f_{yx}$

Classification

- If $A < 0$ and $D > 0$: strict local maximum.
- If $A > 0$ and $D > 0$: strict local minimum.
- If $D < 0$: saddle point.
- If $D = 0$: inconclusive.

Second-order condition

Recap: Concavity, convexity, and global extrema

Convex set

A set S in the x-y plane is convex if, for any two points in S, the full line segment between them also lies in S.

Sufficient conditions on a convex set

For a C^2 function $f(x, y)$ defined on a convex set S:

- If $f_{xx} \leq 0$, $f_{yy} \leq 0$, and $f_{xx}f_{yy} - f_{xy}^2 \geq 0$ throughout S, then f is concave.
- If $f_{xx} \geq 0$, $f_{yy} \geq 0$, and $f_{xx}f_{yy} - f_{xy}^2 \geq 0$ throughout S, then f is convex.

A stationary point of a concave function is a global maximum; a stationary point of a convex function is a global minimum.

Find global extreme

Recap: Extreme Value Theorem in two variables

Key definitions

- A set is **closed** if it contains all its boundary points.
- A set is **bounded** if it fits inside a sufficiently large circle.
- A set that is both closed and bounded is **compact**.

Extreme Value Theorem

If $f(x, y)$ is continuous on a nonempty, closed and bounded set S, then f has both a global maximum and a global minimum on S.

How to find global extrema

Check all interior stationary points and all boundary candidates, then compare function values.

Tutorial Q1

Question 1

Find the first- and second-order partial derivatives.

(a) $z = x^3y^2$

(b) $z = x^5 - 3x^2y + y^6$

(c) $z = \dfrac{x}{y}$

(b) $z = x^5 - 3x^2y + y^6 = f(x,y)$

$z_x = 5x^4 - 6xy + 0 \Rightarrow \begin{cases} z_{xx} = 20x^3 - 6y \\ z_{xy} = -6x \end{cases}$

$z_y = -3x^2 + 6y^5 \Rightarrow \begin{cases} z_{yx} = -6x \\ z_{yy} = 30y^4 \end{cases}$

(c) $z = \dfrac{x}{y}$

$z_x = \dfrac{1}{y} \Rightarrow \begin{cases} z_{xx} = 0 \\ z_{xy} = -\dfrac{1}{y^2} \end{cases}$

$z_y = -\dfrac{x}{y^2} \Rightarrow \begin{cases} z_{yx} = \dfrac{-1}{y^2} \\ z_{yy} = \dfrac{2x}{y^3} \end{cases}$

Tutorial Q1(a): Partial derivatives of $z = x^3y^2$

Knowledge used

- Take the derivative with respect to one variable while holding the other variable constant.
- Compute higher-order partial derivatives by differentiating the first partial derivatives again.

Solution

$$z = x^3y^2.$$

$$z_x = \frac{\partial z}{\partial x} = 3x^2y^2, \qquad z_y = \frac{\partial z}{\partial y} = 2x^3y.$$

$$z_{xx} = \frac{\partial^2 z}{\partial x^2} = 6xy^2, \qquad z_{yy} = \frac{\partial^2 z}{\partial y^2} = 2x^3, \qquad z_{xy} = z_{yx} = 6x^2y.$$

Tutorial Q1(b): Partial derivatives of $z = x^5 - 3x^2y + y^6$

Knowledge used

- Apply ordinary differentiation rules to one variable at a time.
- Treat the other variable as a constant.

Solution

$$z = x^5 - 3x^2y + y^6.$$

$$z_x = 5x^4 - 6xy, \qquad z_y = -3x^2 + 6y^5.$$

$$z_{xx} = 20x^3 - 6y, \qquad z_{yy} = 30y^4, \qquad z_{xy} = z_{yx} = -6x.$$

Tutorial Q1(c): Partial derivatives of $z = x/y$

Knowledge used

- Rewrite $z = x/y$ as $z = xy^{-1}$ when differentiating with respect to y.
- The mixed partials agree when the derivatives are well-defined and continuous.

Solution

$$z = \frac{x}{y}.$$

$$z_x = \frac{1}{y}, \qquad z_y = -\frac{x}{y^2}.$$

$$z_{xx} = 0, \qquad z_{yy} = \frac{2x}{y^3}, \qquad z_{xy} = z_{yx} = -\frac{1}{y^2}.$$

Tutorial Q2

Question 2

For

5 min

$$f(x, y) = x^2 + 2xy^2 + 2y^2,$$

(a) find the first- and second-order partial derivatives;

(b) find and classify the stationary points.

$f(x,y) = x^2 + 2xy^2 + 2y^2$

(a)

$f_x = 2x + 2y^2 \Rightarrow \begin{cases} f_{xx} = 2 \\ f_{xy} = 4y \end{cases}$

$f_y = 4xy + 4y \Rightarrow \begin{cases} f_{yx} = 4y \\ f_{yy} = 4x + 4 \end{cases}$

(b) stationary points: $\begin{cases} f_x = 0 \\ f_y = 0 \end{cases}$

$\Rightarrow \begin{cases} 2x + 2y^2 = 0 \\ 4xy + 4y = 0 \end{cases} \Rightarrow \begin{cases} x + y^2 = 0 & ① \\ xy + y = 0 & ② \end{cases}$

From ①: $x = -y^2$ ③

Plug ③ into ②: $-y^3 + y = 0$

$y(y^2 - 1) = 0 \Rightarrow y = -1, 0, 1$

From ③: $x = -y^2 = \begin{cases} 0, & \text{if } y = 0 \\ -1, & \text{if } y = \pm 1 \end{cases}$

Collect them, $(0,0), (-1,-1), (-1,1)$

Classification

Second-order condition

- If $A < 0$ and $D > 0$: strict local maximum.
- If $A > 0$ and $D > 0$: strict local minimum.
- If $D < 0$: saddle point.
- If $D = 0$: inconclusive.

$A = f_{xx}$, $D = f_{xx} \cdot f_{yy} - f_{xy}^2$

At $(0,0)$: local min

$f_{xx} = 2 > 0$, $f_{xy} = 4y = 0$, $f_{yy} = 4 \times 1 + 4 = 4$

$D = 2 \times 4 - 0 = 8 > 0$

At $(-1,-1)$: Saddle point

$f_{xx} = 2 > 0$, $f_{xy} = 4y = -4$, $f_{yy} = 4 \times 1 + 4 = 0$

$D = 2 \times 0 - (-4) \cdot (-4) = -16 < 0$

At $(-1,1)$: Saddle point

check and verify!

Tutorial Q2(a): First and second partial derivatives

Knowledge used

- Compute f_1 and f_2 by differentiating with respect to the first and second arguments.
- Compute f_{11}, f_{22}, and f_{12} for the second-order test.

Solution

$$f(x, y) = x^2 + 2xy^2 + 2y^2.$$

$$f_1 = 2x + 2y^2, \qquad f_2 = 4xy + 4y.$$

$$f_{11} = 2, \qquad f_{22} = 4x + 4, \qquad f_{12} = 4y.$$

Tutorial Q2(b): Stationary points

Knowledge used

- Interior stationary points satisfy $f_1 = 0$ and $f_2 = 0$.
- Solve the first-order conditions simultaneously.

Solution

The first-order conditions are

$$2x + 2y^2 = 0, \qquad 4xy + 4y = 0.$$

From the second condition,

$$4y(x+1) = 0,$$

so $y = 0$ or $x = -1$.

$$\text{If } y = 0,\ x = 0. \qquad \text{If } x = -1,\ y^2 = 1.$$

Hence the stationary points are

$$(0,0),\ (-1,-1),\ (-1,1).$$

Tutorial Q2(b): Classifying the stationary points

Knowledge used

- Let $A = f_{11}$, $B = f_{12}$, $C = f_{22}$, and $D = AC - B^2$.
- Use the second-order conditions for local extrema.

Solution

$$A = 2, \qquad B = 4y, \qquad C = 4x + 4, \qquad D = AC - B^2.$$

(x_0, y_0)	A	B	C	Result
$(0, 0)$	2	0	4	$D = 8 > 0$, $A > 0$: local min
$(-1, -1)$	2	-4	0	$D = -16 < 0$: saddle point
$(-1, 1)$	2	4	0	$D = -16 < 0$: saddle point

Tutorial Q3

Sufficient conditions on a convex set

For a C^2 function $f(x, y)$ defined on a convex $\boxed{\text{set } S}$:

- If $f_{xx} \le 0$, $f_{yy} \le 0$, and $f_{xx}f_{yy} - f_{xy}^2 \ge 0$ throughout S, then f is concave.
- If $f_{xx} \ge 0$, $f_{yy} \ge 0$, and $f_{xx}f_{yy} - f_{xy}^2 \ge 0$ throughout S, then f is convex.

A stationary point of a concave function is a global maximum; a stationary point of a convex function is a global minimum.

Question 3

A firm sells in two markets. The demand equations are

$$0.1p_1 - 1.2 + 0.2q_1 = 0, \qquad 10p_2 - 320 + 40q_2 = 0.$$

p_2

The cost function is $C = q_1^2 + q_2^2 + 2q_1q_2$. Find the profit-maximising quantities and prices.

profit: π = Revenue − cost

$= p \cdot q - C$

$= p_1 \cdot q_1 + p_2 \cdot q_2 - (q_1^2 + q_2^2 + 2q_1q_2)$ ①

From demands: $p_1 = 12 - 2q_1$ ②

$p_2 = 32 - 4q_2$ ③

plug ②, ③ into ①:

$\pi = (12 - 2q_1) \cdot q_1 + (32 - 4q_2) \cdot q_2 - (q_1^2 + q_2^2 + 2q_1q_2)$

$= 12q_1 + 32q_2 - 3q_1^2 - 5q_2^2 - 2q_1q_2$

$= \pi(q_1, q_2)$

$\pi_1 = 12 - 6q_1 - 2q_2$, $\pi_2 = 32 - 10q_2 - 2q_1$

Stationary point $(1,3)$: $\begin{cases} \pi_1 = 12 - 6q_1 - 2q_2 = 0 \\ \pi_2 = 32 - 10q_2 - 2q_1 = 0 \end{cases} \Rightarrow \begin{cases} q_1 = 1 \\ q_2 = 3 \end{cases}$

$\pi_{11} = -6$, $\pi_{12} = -2$, $\pi_{21} = -2$, $\pi_{22} = -10$

$D = \pi_{11} \cdot \pi_{22} - \pi_{12}^2 = 56 > 0$

$\Rightarrow \pi_{11} < 0$, $\pi_{22} < 0$, $D > 0$ for all (q_1, q_2)

$\Rightarrow \pi$ is concave

$\Rightarrow$ Stationary point $(q_1, q_2) = (1, 3)$ is a global max

$\Rightarrow$ plug $(1,3)$ into ②, ③ $\Rightarrow$ price

Tutorial Q3: Profit function

Knowledge used

- Rewrite demand equations as inverse demand functions.
- Profit equals total revenue minus total cost.

Solution

From the demand equations,

$$p_1 = 12 - 2q_1, \qquad p_2 = 32 - 4q_2.$$

Profit is

$$\pi(q_1, q_2) = p_1 q_1 + p_2 q_2 - C.$$

Therefore

$$\pi = (12 - 2q_1)q_1 + (32 - 4q_2)q_2 - q_1^2 - q_2^2 - 2q_1 q_2,$$

so

$$\pi(q_1, q_2) = 12q_1 + 32q_2 - 3q_1^2 - 5q_2^2 - 2q_1 q_2.$$

Tutorial Q3: First-order conditions

Knowledge used

- For an interior optimum, solve $\pi_1 = 0$ and $\pi_2 = 0$.
- These are simultaneous equations in q_1 and q_2.

Solution

$$\frac{\partial \pi}{\partial q_1} = 12 - 6q_1 - 2q_2, \qquad \frac{\partial \pi}{\partial q_2} = 32 - 10q_2 - 2q_1.$$

Set both equal to zero:

$$12 - 6q_1 - 2q_2 = 0, \qquad 32 - 10q_2 - 2q_1 = 0.$$

From the first equation,

$$q_2 = 6 - 3q_1.$$

Substitute into the second:

$$32 - 10(6 - 3q_1) - 2q_1 = 0 \Rightarrow q_1 = 1, \qquad q_2 = 3.$$

Tutorial Q3: Second-order conditions and prices

Knowledge used

- Use the Hessian matrix to check concavity.
- A concave profit function has a global maximum at its stationary point.

Solution

$$\boldsymbol{H}_\pi = \begin{pmatrix} -6 & -2 \\ -2 & -10 \end{pmatrix}.$$

Here

$$A = -6 < 0, \qquad C = -10 < 0, \qquad AC - B^2 = (-6)(-10) - (-2)^2 = 56 > 0.$$

Thus profit is concave and the stationary point is a global maximum.

$$q_1^* = 1, \quad q_2^* = 3.$$

Prices:

$$p_1 = 12 - 2(1) = 10, \qquad p_2 = 32 - 4(3) = 20.$$

Tutorial Q4

Question 4

Demand for butter and margarine is given by

B M

$$Q_b = 8000 - 0.09P_b^2 + 0.008P_m^2,$$

$$Q_m = 5000 + 0.004P_b^2 - 0.03P_m^2.$$

Use partial derivatives to determine whether butter and margarine are substitutes or complements.

Cross-price effect

$$\frac{\partial Q_b}{\partial P_m} = \frac{\partial}{\partial P_m}(8000 - 0.09P_b^2 + 0.008P_m^2) = 0.016P_m > 0$$

$$\frac{\partial Q_m}{\partial P_b} = \frac{\partial}{\partial P_b}(5000 + 0.004P_b^2 - 0.03P_m^2) = 0.008P_b > 0$$

B and M are substitutes !

Cross-price effects

If demand for good A depends on the price of good B, then

$$\boxed{\frac{\partial q_A}{\partial p_B}}$$

measures how demand for A responds to the price of B.

Substitutes vs complements

- If $\frac{\partial q_A}{\partial p_B} > 0$, then A and B are substitutes.
- If $\frac{\partial q_A}{\partial p_B} < 0$, then A and B are complements.
- If $\frac{\partial q_A}{\partial p_B} = 0$, there is no cross-price effect.

Tutorial Q4: Cross-price effects

Knowledge used

- Cross-price partial derivatives measure how demand for one good responds to the price of another good.
- A positive cross-price effect suggests substitutes; a negative effect suggests complements.

Solution

For butter demand,

$$\frac{\partial Q_b}{\partial P_m} = 0.016 P_m.$$

For margarine demand,

$$\frac{\partial Q_m}{\partial P_b} = 0.008 P_b.$$

Tutorial Q4: Cross-price effects

Knowledge used

- Cross-price partial derivatives measure how demand for one good responds to the price of another good.
- A positive cross-price effect suggests substitutes; a negative effect suggests complements.

Solution

Since prices are nonnegative, these partial derivatives are nonnegative:

$$\frac{\partial Q_b}{\partial P_m} \geq 0, \qquad \frac{\partial Q_m}{\partial P_b} \geq 0.$$

Therefore, when the price of one good increases, demand for the other good increases. Hence butter and margarine are

substitutes.

Key takeaways

- Partial derivatives measure how a function changes with one variable, holding the others constant.
- Multivariate stationary points solve all first-order partial derivative equations at once.
- The Hessian matrix helps classify stationary points as local maxima, local minima, or saddle points.
- Cross-price partial derivatives help identify substitutes and complements.